

# Oleksandr Kolko

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## PROFESSIONAL SUMMARY

BSc Cybersecurity and Forensic Computing graduate from the University of Portsmouth, finishing in July 2026 with First Class Honours. I have hands-on experience building and maintaining containerised infrastructure with Docker and Kubernetes, automating deployment workflows with Python, and setting up CI/CD pipelines from scratch during a real internship. My cybersecurity background means I approach cloud and infrastructure work with security built in rather than added on. I am looking to build a career in cloud or infrastructure engineering where I can develop across AWS, GCP and Azure and work on complex technical problems from day one.

## CORE COMPETENCIES

AWS

Terraform

Docker

Kubernetes

Helm

CI/CD

Python

Linux

IAM & Least Privilege

Networking

## WORK EXPERIENCE

CyberVectors Labs

October 2025 - February 2026

### Cybersecurity Engineering Intern

Remote

- Provisioned and maintained up to 5 concurrent containerised lab environments on AWS using Terraform (ECS and EKS on Fargate, backed by EC2 compute), managing configuration, dynamic port allocation, session isolation and consistent uptime across all running instances.
- Built automated provisioning workflows integrating Python (FastAPI) and PHP (Laravel) backends with Docker APIs and AWS Lambda, reducing manual setup time and improving deployment consistency.
- Configured VPCs, security groups and network ACLs as Terraform-managed infrastructure to isolate lab environments from each other, and applied IAM least-privilege policies so users could only access their allocated resources.
- Managed secrets and credentials through AWS Secrets Manager and KMS rather than storing them in plaintext, and used S3 for lab environment data and artefact storage.
- Set up CI/CD pipelines using AWS CodePipeline, CodeBuild and CodeCommit alongside Git-based version control, standardising how the team deployed and tracked infrastructure changes and ensuring consistent, auditable releases.
- Monitored running environments and used CloudTrail to trace issues back to their root cause, triaging infrastructure problems independently and resolving them in a timely manner.
- Wrote technical documentation, deployment guides and architecture overviews used by both engineers and non-technical stakeholders.

Calvin Klein

June 2024 - Present

### Sales Assistant

Portsmouth

- Assisted 50+ customers daily in a fast-paced environment, adapting communication to resolve complex queries quickly.
- Managed inventory operations, demonstrating attention to detail and reliability under pressure.
- Worked within a team to meet operational targets consistently.

## PROJECTS

### GCP/GKE DevSecOps Platform

Deployed a FastAPI service to Google Kubernetes Engine entirely as code: Terraform-provisioned VPC-native GKE Standard cluster on Spot nodes (autoscaling 1-5), Artifact Registry and GCS-backed remote state. Engineered a keyless CI/CD pipeline in GitHub Actions that gates on pytest, authenticates to GCP via Workload Identity Federation with zero static service-account keys, then builds, SHA-tags and helm upgrades the app on every merge to main. Implemented detection-as-code in Terraform (Cloud Audit Logs to log-based metric to Monitoring alert on service-account-key creation) alongside kube-prometheus-stack and a Grafana dashboard, plus FinOps guardrails with a GBP 10/month billing budget and cost-attribution labels.

Terraform, GKE, Helm, GitHub Actions, Workload Identity Federation, Prometheus, Grafana, Kafka

Anomaly Detection Proxy

Built a Python reverse-proxy anomaly detector inspecting inbound HTTP traffic in real time, flagging rate-limit abuse, path probing and known scanner user-agents, logging and forwarding rather than blocking to avoid disrupting legitimate users. Deployed on Kubernetes with 3 replicas, validated self-healing by deleting pods, and packaged it with Helm using per-environment values files for one-command deploys and reversible releases. Managed AWS credentials as Kubernetes Secrets injected at runtime and persisted alerts to S3 as timestamped JSON.

Python, Kubernetes, Helm, Docker, AWS S3

Machine Learning Phishing Detection Browser Extension Final-year project - 75%

Built a Chrome/Firefox extension detecting phishing websites in real time using locally-run ML models, with no browsing data sent to third-party services. Engineered features from URL structure and page content, then trained and evaluated Random Forest and XGBoost models on approximately 8,000 URLs, achieving an 8% false positive rate. Covered the full development lifecycle independently.

Python, XGBoost, Random Forest, Browser Extensions

Cloud Security Scanner

Built a modular Python security scanner with three independent modules: TCP port scanning, HTTP security header checking and S3 misconfiguration detection. Found and fixed real S3 misconfigurations on my own AWS infrastructure using the scanner output, patching versioning and logging via Terraform rather than the AWS console.

Python, AWS, Terraform, Git

Home Lab - Enterprise Infrastructure Simulation

Architected a 6-VM environment simulating an enterprise network topology with VPN, firewall rules and interconnectivity controls, then simulated network failure scenarios to validate system resilience and recovery protocols.

Virtualisation, VPN, Firewalls, Linux

EDUCATION

BSc Cybersecurity and Forensic Computing (BCS-accredited) University of Portsmouth September 2023 - July 2026

First Class Honours, dissertation graded 75%. Relevant modules: Networking and Operating Systems; Architecture and Operating Systems; IT and Internetworking Security (CCNA Security-aligned); Security Management (ISO 27001, NIST CSF, GDPR); Malware and Vulnerability; Security and Cryptography; Ethical Hacking.

CERTIFICATIONS

|                                          |                     |             |
|------------------------------------------|---------------------|-------------|
| AWS Certified Cloud Practitioner         | Amazon Web Services | In progress |
| CompTIA Security+                        | CompTIA             | In progress |
| CompTIA Network+                         | CompTIA             | In progress |
| TryHackMe - Top 3%, 140+ rooms completed | TryHackMe           | Ongoing     |

SKILLS

**Cloud and Infrastructure:** AWS hands-on across EC2, ECS, EKS, Fargate, Lambda, VPC, security groups, NACLs, S3, IAM, Secrets Manager and KMS, provisioned as Terraform-managed infrastructure during a real internship. CodePipeline, CodeBuild and CodeCommit for CI/CD, with CloudTrail for debugging. Building familiarity with GCP and Azure.

**Containerisation and DevOps:** Extensive hands-on Docker experience. Hands-on Kubernetes: deployments, replica scaling, self-healing, NodePort Services and Secrets, with applications packaged and released via Helm using per-environment values files. Familiar with event-driven messaging (Kafka, NATS). CI/CD pipeline design and Git throughout.

**Scripting and Automation:** Python as primary language for automation, API integration and tooling (FastAPI). Bash and shell scripting. Comfortable writing scripts to reduce manual toil and improve deployment consistency.

**Linux and Networking:** Comfortable on the Linux command line day-to-day across Ubuntu and Alpine. Solid foundations in TCP/IP, DNS, DHCP, VLANs, firewalls and VPCs.

**Security and Compliance:** ISO 27001, NIST CSF and GDPR awareness built through a full cybersecurity degree. Security-aware approach to cloud infrastructure design across network, firewall, OS, logging and monitoring standards.

**Tools:** Wireshark, Nmap, Burp Suite, WPScan, FTK Imager, RegRipper, Splunk, Terraform, Git.